



Investigation of Current-Phase Relationship in 1T-Phase Platinum Ditelluride Josephson Junctions under In-Plane Magnetic Field

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Tao-Yi Hsu (□□□)

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National Sun Yat-sen University

Outline

1. Introduction

- Research Motivation
- Type-II Dirac Semimetal (1T-PtTe₂)

2. Theoretical Background

- Josephson Effect & CPR
- Josephson Diode Effect (JDE)

3. Experimental Methods

- Device Fabrication
- Measurement Setup

4. Results & Discussion

- Basic Characterization
- Current-Phase Relation (CPR) Analysis
- Magnetic Field Dependence

5. Conclusion

Experimental Methods

Device Fabrication Flow

1. **Crystal Growth:** Self-flux method (provided by Prof. C.S. Lue, NCKU).
2. **Exfoliation:** Mechanical exfoliation of PtTe_2 flakes onto SiO_2/Si substrates.
3. **E-beam Lithography (EBL):**
 - PMMA resist coating.
 - Pattern definition for contacts and SQUID loops.
4. **Metallization:**
 - Superconducting electrodes: **Al (Aluminum)** or **Nb (Niobium)**.
 - *In-situ* Ar ion milling to ensure transparent interfaces.
 - E-beam evaporation & Liftoff.
5. **Design:** Asymmetric dc-SQUID geometry for CPR retrieval.

Measurement Setup

- **Cryogenic System:** Dilution Refrigerator (DR)
 - Base temperature: $\sim 10 \text{ mK}$

Results: Basic Characterization

I-V Characteristics (DC Transport)

- **Typical IV Curve:**

- Clear Superconducting branch ($V = 0$).
- Switching Current (I_c) and Retrapping Current (I_r).
- Hysteretic behavior (underdamped junction).

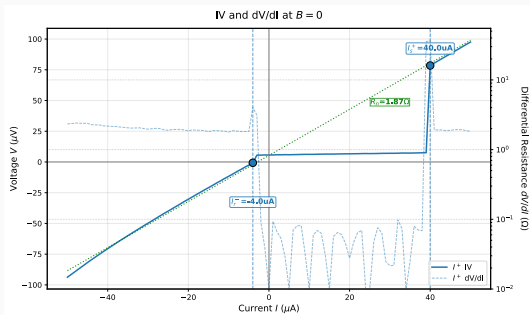


Figure 3: I-V Curve at Base Temp.

- **Observation:**

- Sharp transition indicates high-quality junction.
- Finite resistance in normal state (R_N).
- **Excess Current** observed at high bias $\rightarrow$ High transparency interface.

Results: Current-Phase Relation

CPR Retrieval Method

- **Asymmetric SQUID Technique:**

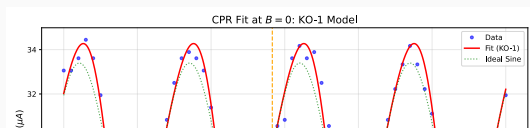
- Large junction (I_{ref}) // Small junction (I_{test}).
- $I_c(\Phi_{ext}) \approx I_{ref} + I_{test}(\phi)$.
- Assumes $I_{ref} \gg I_{test}$ and sinusoidal $I_{ref}(\phi)$.

- **Math Model:**

$$I_s(\phi) = I_1 \sin(\phi) + I_2 \sin(2\phi + \delta)$$

- I_1 : Fundamental component (conventional).
- I_2 : Second harmonic (4π -periodic / ABS).

Measured CPR & Fitting



- **Key Findings:**

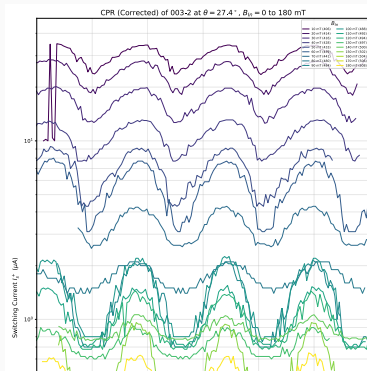
- **Skewed CPR:** Deviation from pure sine wave.
- **High I_2 / I_1 ratio:** Significant

Results: In-Plane Magnetic Field Effect

Tunable CPR with In-Plane Field

- **Observation:**

- Applying $B_{||}$ modifies the CPR shape.
- Induces a **phase shift** (ϕ_0) in the second harmonic?
- Evidence of **Josephson Diode Effect (JDE)**.



Conclusion

Summary

1. **Successful Fabrication:**

- High-quality 1T-PtTe₂ Josephson junctions and SQUIDs realized.

2. **Anomalous CPR:**

- Observed significant **second harmonic** (I_2) contribution.
- Direct evidence of high-transparency transport modes.

3. **Magnetic Tuning:**

- In-plane magnetic field effectively modulates the CPR and induces **Josephson Diode Effect**.
- Demonstrates potential for **tunable superconducting logic** and **phase batteries**.

Future Outlook

- **Optimization:** Improving fabrication yield and interface control.
- **Material Exploration:** Extending to other Weyl/Dirac semimetals.
- **Quantum Information:** Integration into superconducting qubit circuits (e.g., as

References